

Problem-Solution Fit

Date	30 June
Project Name	Rising Waters
Maximum Marks	5 Marks

Problem-Solution Fit Canvas:

Use the framework below to ensure your proposed solution accurately addresses the root causes, behaviors, and limitations of your target audience. Fill out the canvas in the respective boxes just like the original visual template.

1. CUSTOMER SEGMENT(S) [CS] Regional disaster management teams, municipal flood response coordinators, and local emergency dispatch personnel.	6. CUSTOMER LIMITATIONS [CL] Limited bandwidth in severe storm environments, restricted field budgets, and reliance on mid-range mobile terminal devices.	5. AVAILABLE SOLUTIONS [AS] Pros: Standard radar weather tracking maps offer broad regional forecasts. Cons: Lacks neighborhood stream gauge resolution and real-time operational alerts.
2. PROBLEMS / PAINS + FREQUENCY [PR] Delayed warnings on localized creek flash floods. Occurs multiple times during every active monsoon window.	9. PROBLEM ROOT / CAUSE [RC] Siloed structural telemetry endpoints and a lack of unified predictive alerts mapping localized rainfall to specific gauge thresholds.	7. BEHAVIOR + INTENSITY [BE] Frequent manual checking of multiple disjointed weather applications and public gauges during severe storms; High stress intensity.
3. TRIGGERS TO ACT [TR] Precipitation hitting initial threshold bounds or upstream telemetry indicating abnormal surging patterns.	10. YOUR SOLUTION [SL] A unified Flask data visualization dashboard connecting sensor APIs with localized classification boundaries and automated Twilio SMS alert escalations.	8. CHANNELS OF BEHAVIOR [CH] ONLINE: Dynamic dashboard cards, live map visualization tools, and SMS notifications. OFFLINE: Manual field radio cross-checks and physical site check verification tools.
4. EMOTIONS BEFORE / AFTER [EM] Before: Anxious, overwhelmed by fragmented reports. After: Confident, precise, and proactive due to centralized monitoring metrics.		